

Supplementary Material: Radical Solvability and Numerical Validation of the Scalar ℓ_p Proximal Equation

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1 Radical Solvability of the Scalar ℓ_p Proximal Equation

The main paper shows that for rational $p \in (0, 1)$ the series representation of the larger proximal root can be reorganized into a finite sum of generalized hypergeometric functions, yielding explicit closed-form representations for several rational exponents, including $p = \frac{1}{3}, \frac{1}{2}, \frac{2}{3}$. A different and more restrictive question is whether the scalar proximal equation admits a *uniform formula by radicals*. The purpose of this supplement is to answer that question in the negative, showing that among rational exponents $p \in (0, 1)$, only $p = \frac{1}{2}$ and $p = \frac{2}{3}$ admit such a formula.

The Abel–Ruffini theorem [3, 4] states that there is no general formula by radicals for polynomial equations of degree at least five. In modern Galois-theoretic terms, this reflects the fact that the generic polynomial of degree n has Galois group S_n , and S_n is not solvable for $n \geq 5$. This suggests that, once the polynomial associated with the scalar ℓ_p proximal equation has generic Galois group S_n with $n \geq 5$, one should not expect a uniform radical formula.

We consider the scalar proximal problem

$$\text{prox}_{\Lambda|\cdot|^p}(y) := \arg \min_{x \in \mathbb{R}} \left\{ \frac{1}{2}(x - y)^2 + \Lambda|x|^p \right\}, \quad 0 < p < 1, \quad \Lambda > 0. \quad (1)$$

By symmetry, it suffices to work with the nonnegative parameter

$$Z := |y| \geq 0.$$

On the nonzero branch $x > 0$, the critical-point equation becomes

$$x - Z + \Lambda p x^{p-1} = 0. \quad (2)$$

We treat Z and Λ as algebraically independent indeterminates over $\mathbb{R}$ and ask whether the nonzero solution of (2) can be expressed by radicals over the field $\mathbb{R}(Z, \Lambda)$, the rational function field in Z and Λ over $\mathbb{R}$.

Definition 1.1. Let K be a field of characteristic 0. An element α is said to be *expressible by radicals over K* if there exists a tower of field extensions

$$K = K_0 \subset K_1 \subset \cdots \subset K_r$$

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such that, for each $i = 1, \dots, r$,

$$K_i = K_{i-1}(\beta_i), \quad \beta_i^{d_i} \in K_{i-1}$$

for some integer $d_i \geq 2$, and $\alpha \in K_r$. Such an extension K_r/K is called a *radical extension* of K .

A polynomial $f(x) \in K[x]$ is said to be *solvable by radicals over K* if all of its roots belong to some radical extension of K . Equivalently, the splitting field of f over K is contained in a radical extension of K .

For background on radical extensions and solvability by radicals, see [3, 4].

The following classical lemma is the key Galois-theoretic input.

Lemma 1.2 (Generic trinomial Galois group). *Let F be a field of characteristic 0, and let A, B be algebraically independent over F . Let $N > M \geq 1$ be integers with $\gcd(N, M) = 1$. Then the trinomial*

$$T^N + AT^M + B \in F(A, B)[T]$$

has Galois group S_N over $F(A, B)$.

For $M \geq 2$, this follows from Smith [2]; for $M = 1$, see Uchida [5]. See also Osada [1] for a related treatment of trinomials of the form $X^n + aX + b$.

In Lemma 1.2, the conclusion that the Galois group is S_N means that the trinomial has the full symmetric group as its Galois group, so its roots admit the largest possible permutation symmetry. Since S_N is not solvable for $N \geq 5$, this immediately rules out solvability by radicals in those cases.

We now obtain the classification relevant to the scalar ℓ_p proximal equation.

Theorem 1.3. *Let*

$$p = \frac{m}{n} \in (0, 1) \cap \mathbb{Q}, \quad \gcd(m, n) = 1, \quad 0 < m < n.$$

Then the nonzero branch of the scalar ℓ_p proximal equation (2) admits a uniform radical formula in the symbolic parameters (Z, Λ) if and only if

$$p = \frac{1}{2} \quad \text{or} \quad p = \frac{2}{3}.$$

Proof. Write $p = \frac{m}{n}$ in lowest terms and introduce the substitution

$$x = t^n.$$

Then (2) becomes

$$t^n - Z + \Lambda \frac{m}{n} t^{m-n} = 0.$$

Multiplying by t^{n-m} yields the trinomial equation

$$t^{2n-m} - Zt^{n-m} + \frac{m}{n}\Lambda = 0. \tag{3}$$

Set

$$N := 2n - m, \quad M := n - m, \quad C := \frac{m}{n}\Lambda.$$

Then (3) can be written as

$$f_{m,n}(T) = T^N - ZT^M + C.$$

Moreover,

$$\gcd(N, M) = \gcd(2n - m, n - m) = \gcd(n, m) = 1.$$

We first show that radical solvability for x is equivalent to radical solvability for t over $\mathbb{R}(Z, \Lambda)$. Since $x = t^n$, any radical expression for t immediately gives one for x . Conversely, from (3) we obtain

$$t^{n-m} = -\frac{C}{x-Z}.$$

Because $\gcd(n, n-m) = \gcd(n, m) = 1$, there exist integers a, b such that

$$an + b(n-m) = 1.$$

Hence

$$t = t^{an+b(n-m)} = (t^n)^a (t^{n-m})^b = x^a \left(-\frac{C}{x-Z} \right)^b,$$

which shows that

$$t \in \mathbb{R}(Z, \Lambda)(x).$$

Therefore, x is expressible by radicals over $\mathbb{R}(Z, \Lambda)$ if and only if t is.

Now apply Lemma 1.2 to $f_{m,n}(T)$ with

$$A = -Z, \quad B = C = \frac{m}{n}\Lambda.$$

Since Z and Λ are algebraically independent over $\mathbb{R}$, so are $-Z$ and C . Therefore

$$\text{Gal}(f_{m,n}/\mathbb{R}(Z, \Lambda)) \cong S_N = S_{2n-m}.$$

In particular, $f_{m,n}$ is irreducible, so it is the minimal polynomial of the generic root t . If t belonged to a radical extension, then after adjoining the required roots of unity and taking the normal closure, all conjugates of t would lie in a solvable Galois extension. The splitting field of $f_{m,n}$ would therefore have a solvable Galois group. Hence a uniform radical formula for (2) would force S_N to be solvable. Since this holds if and only if $N \leq 4$, a necessary condition is

$$2n - m \leq 4.$$

We now solve this arithmetic constraint under

$$0 < m < n, \quad \gcd(m, n) = 1.$$

If $n \geq 4$, then

$$2n - m \geq n + 1 \geq 5,$$

which is impossible. Hence only $n = 2$ or $n = 3$ can occur.

If $n = 2$, then necessarily $m = 1$, so $p = \frac{1}{2}$. If $n = 3$, then $m = 1$ or $m = 2$. The choice $m = 1$ gives $2n - m = 5$, which is impossible, whereas $m = 2$ yields $p = \frac{2}{3}$. Thus the only possible rational exponents are

$$p = \frac{1}{2} \quad \text{and} \quad p = \frac{2}{3}.$$

It remains to prove sufficiency. If $p = \frac{1}{2}$, then $x = t^2$ and (3) becomes

$$t^3 - Zt + \frac{\Lambda}{2} = 0,$$

which is a cubic equation and therefore solvable by radicals via Cardano's formula. If $p = \frac{2}{3}$, then $x = t^3$ and (3) becomes

$$t^4 - Zt + \frac{2\Lambda}{3} = 0,$$

which is a quartic equation and therefore solvable by radicals via Ferrari's formula. Hence the nonzero branch of (2) admits a uniform radical formula in the symbolic parameters (Z, Λ) if and only if

$$p = \frac{1}{2} \quad \text{or} \quad p = \frac{2}{3}.$$

□

Remark 1.4. Theorem 1.3 concerns *solvability by radicals*, not the broader existence of special-function representations. Accordingly, there is no contradiction with the hypergeometric formulas in the main paper, including the case $p = \frac{1}{3}$. The theorem only says that among rational exponents in $(0, 1)$, the scalar ℓ_p proximal equation admits a uniform formula by radicals only for $p = \frac{1}{2}$ and $p = \frac{2}{3}$. It also does not exclude accidental solvability for special numerical choices of $(|y|, \Lambda)$, where the specialized polynomial may have a smaller Galois group.

2 Additional numerical validation

These experiments illustrate the accuracy of the series, the separation of contour and quadrature errors, and the acceptance conditions for inexact proximal-gradient steps. We retain the notation of the main paper and write $z = x_+(y)/y$. For scalar tests, the reported error is $e_z = |\hat{z} - z_{\text{ref}}|$, equivalently $|\hat{x} - x_{\text{ref}}|/y$. Reference values are obtained independently by monotone bisection at 70 and 100 decimal digits. The reference problem uses the exact binary floating-point inputs supplied to the evaluators. Evaluator outputs are stored in double precision; stopping residuals use extended precision where available. The results are numerical checks, not validated interval certificates.

2.1 Accuracy across the active domain

We consider the 135 combinations

$$\begin{aligned} p &\in \{0.1, \tfrac{1}{3}, \tfrac{1}{2}, \tfrac{2}{3}, 0.8, 0.9, 0.95, 0.99, 0.999\}, \\ \lambda &\in \{10^{-6}, 0.8, 10^6\}, \\ y/\tau_{p,\lambda} - 1 &\in \{10^{-10}, 10^{-6}, 10^{-2}, 0.5, 9\}. \end{aligned}$$

The coefficients and powers are combined in the logarithmic domain before summation. Besides fixed orders $N = 10, 50, 110$, we use an adaptive series that doubles N , starting from 16, up to a limit of 16384. Its stopping indicator is

$$\frac{|\hat{z} - 1 + \varepsilon \hat{z}^{p-1}|}{1 - p/2}, \quad \hat{z} \in [\rho_{p,\lambda}/y, 1],$$

with target 10^{-11} . By the active-branch derivative bound and the mean-value theorem, the indicator bounds e_z in exact arithmetic whenever the candidate lies in the stated interval. Double-precision bisection and safeguarded Newton provide additional reference implementations. All outputs are checked against the independent high-precision root.

The adaptive series meets the target in 129 cases. The remaining six cases have $p = 0.999$ and relative activation gaps 10^{-10} or 10^{-6} , for each of the three penalties. Bisection and Newton meet

Table 1: Observed normalized accuracy on the 135 scalar problems. “Accurate” means $e_z \leq 10^{-11}$; the target is used only for reporting for fixed-order sums. The largest error includes cases that reached the adaptive work limit.

Evaluator	Accurate / total	Largest e_z	Caps
Series, $N = 10$	30 / 135	2.13×10^{-2}	–
Series, $N = 50$	72 / 135	4.63×10^{-3}	–
Series, $N = 110$	90 / 135	2.05×10^{-3}	–
Residual-controlled series	129 / 135	1.00×10^{-7}	6
Bisection	135 / 135	8.82×10^{-12}	0
Safeguarded Newton	135 / 135	6.79×10^{-12}	0

the target in all cases. Figure 1 distinguishes the slow series convergence from the conditioning of the active root. In particular, the computed derivative stays below $(1 - p/2)^{-1}$, consistently with the uniform conditioning result in the main paper. The deterioration as $p \uparrow 1$ is consistent with $q_p = p2^{1-p} \uparrow 1$; the uniform series-convergence statement fixes p .

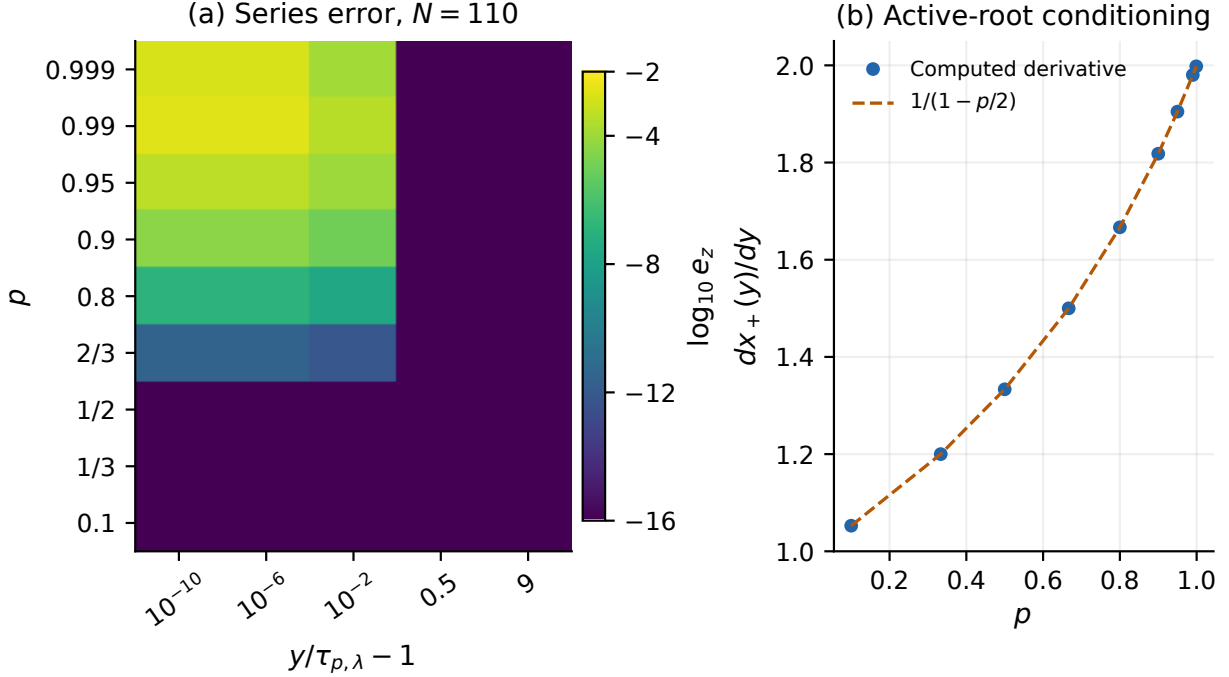


Figure 1: Accuracy and conditioning for $\lambda = 0.8$. Left: normalized error of the series with $N = 110$; the color scale saturates below 10^{-16} and above 10^{-2} . Right: the derivative evaluated at the reference root with $y/\tau_{p,\lambda} - 1 = 10^{-10}$, together with its theoretical upper bound.

2.2 Contour truncation and quadrature error

We evaluate the residue-corrected formula using

$$H(s) = -\frac{\Gamma(-s)\Gamma((2-p)s-1)}{\Gamma((1-p)s)}, \quad \sigma_N = N + \frac{1}{2},$$

and the boundary value $\text{Log}(-\varepsilon) = \log \varepsilon + i\pi$. The finite vertical integral is approximated by the composite trapezoidal rule with spacing $h = 2T/M$. Gamma factors and the complex power are combined in the logarithmic domain. Errors use the full complex norm; the imaginary component is retained in the accompanying data.

The contour study fixes $\lambda = 0.8$ and varies $p \in \{\frac{1}{2}, \frac{2}{3}, 0.9, 0.99\}$, $y/\tau_{p,\lambda} \in \{1 + 10^{-6}, 1.5\}$, $N \in \{10, 30\}$, $T \in \{10, 50, 200\}$, and $h \in \{0.5, 0.1, 0.02\}$, giving 144 evaluations. Figure 2 shows representative near-activation cases at the finest spacing. Their errors decrease over the tested heights; the larger exponent requires greater truncation effort.

These experiments demonstrate the separation of error sources, not uniform speed superiority of the contour formula. Near $p = 1$, the required contour height and quadrature cost increase, so monotone bisection or safeguarded Newton remains the practical fallback.

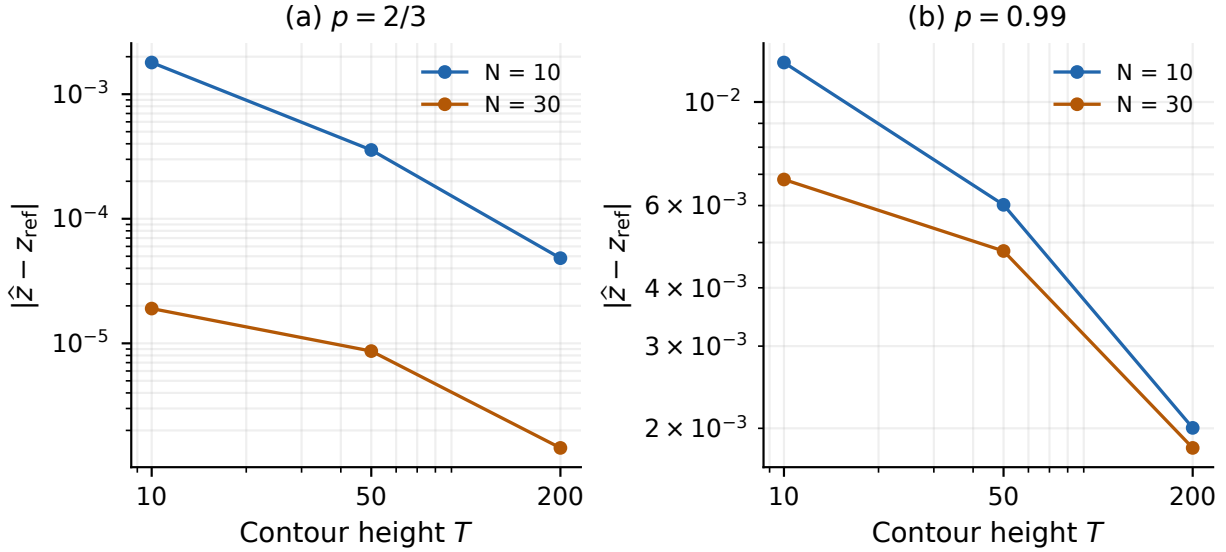


Figure 2: Residue-corrected errors for $\lambda = 0.8$, $y/\tau_{p,\lambda} = 1 + 10^{-6}$, and $h = 0.02$. The curves show the full complex error for two truncation orders. These finite-height observations do not identify an asymptotic convergence rate.

For an independent error decomposition, the finite integral is recomputed at 40 and 60 decimal digits, while the high-precision quadrature panel width is halved from 2 to 1. We use $(p, y, N, T) = (\frac{2}{3}, 1.26, 10, 5)$, $(\frac{2}{3}, 1.50, 10, 20)$, and $(\frac{1}{2}, 1.50, 3, 10)$, with $\lambda = 0.8$. The two high-precision calculations agree within 4.2×10^{-38} . The measured components satisfy the triangle decomposition

$$e_z \leq E_{\text{tail}} + E_{\text{int}} + E_{\text{arith}},$$

where E_{tail} is the finite-contour remainder, E_{int} is the finite-integral evaluation error, and E_{arith} includes series evaluation, input normalization, and final addition. These components are evaluated a posteriori.

In Table 2, each fourfold mesh refinement reduces E_{int} by approximately a factor of 16, while the total error approaches the finite-contour remainder. This illustrates the distinction between contour truncation and numerical quadrature in the error analysis. The decomposition complements the analytic bounds; its components remain empirical high-precision measurements.

Table 2: Measured error components for $p = \frac{2}{3}$, $\lambda = 0.8$, $y = 1.26$, $N = 10$, and $T = 5$. The combined arithmetic component is below 1.4×10^{-16} in these three evaluations.

	h	E_{int}	E_{tail}	e_z
	0.2	5.71×10^{-7}	1.92×10^{-3}	1.92×10^{-3}
	0.05	3.57×10^{-8}	1.92×10^{-3}	1.92×10^{-3}
	0.0125	2.23×10^{-9}	1.92×10^{-3}	1.92×10^{-3}

2.3 Residual-controlled proximal-gradient steps

We consider

$$F(x) = \frac{1}{2}\|Ax - b\|^2 + \lambda\|x\|_p^p.$$

Two Gaussian matrices $A \in \mathbb{R}^{40 \times 70}$ are generated and their columns normalized. Each ground truth has six randomly selected nonzero Gaussian entries, and $b = Ax^{\text{true}} + 0.01\xi$, with standard Gaussian noise ξ . Seeds 20260915 and 20260916 define the two instances. For each instance and $p \in \{0.5, 0.8\}$, the series and safeguarded-Newton proximal evaluations share the same data, $x_0 = 0$, $\lambda = 0.04$, and $\alpha = 0.9/\|A\|_2^2$.

Steps are accepted using the model-decrease and residual conditions of the main paper, with

$$\delta_k = \frac{10^{-12}}{(k+1)^2}, \quad \eta_k = \frac{10^{-4}}{(k+1)^{1.25}}.$$

The model difference and residual are checked directly at the computed point. A Newton fallback is available if the scalar series exceeds 4096 terms; no fallback was used. We allow at most 200 steps and stop when

$$\frac{\|x_{k+1} - x_k\|}{\max\{1, \|x_k\|\}} \leq 10^{-8}.$$

The stationarity measure is

$$S(x) = \left\| \left((A^\top(Ax - b))_i + \lambda p \operatorname{sgn}(x_i)|x_i|^{p-1} \right)_{i:x_i \neq 0} \right\|_2.$$

Table 3: Terminal values for the two shared instances. S and N denote series and safeguarded-Newton proximal evaluations. Seeds 1 and 2 correspond to 20260915 and 20260916.

Seed	p	Oracle	K	$F(x_K)$	$S(x_K)$
1	0.5	S	177	0.239049	1.62×10^{-7}
1	0.5	N	177	0.239049	1.62×10^{-7}
1	0.8	S	190	0.248428	1.76×10^{-7}
1	0.8	N	190	0.248428	1.76×10^{-7}
2	0.5	S	198	0.263414	1.06×10^{-7}
2	0.5	N	198	0.263414	1.06×10^{-7}
2	0.8	S	188	0.200700	1.09×10^{-7}
2	0.8	N	188	0.200700	1.09×10^{-7}

All eight runs satisfy the stopping condition within 177–198 iterations. Across their 1506 accepted steps, the model-decrease condition, residual condition, and predicted descent inequality all hold without the recorded roundoff allowances. The largest ratio $\|r_{k+1}\|/\eta_k$ is approximately

2.31×10^{-2} . Table 3 and Figure 3 show the objective and stationarity behavior. These finite runs illustrate that the theorem's acceptance conditions can be implemented and are consistent with its asymptotic conclusions.

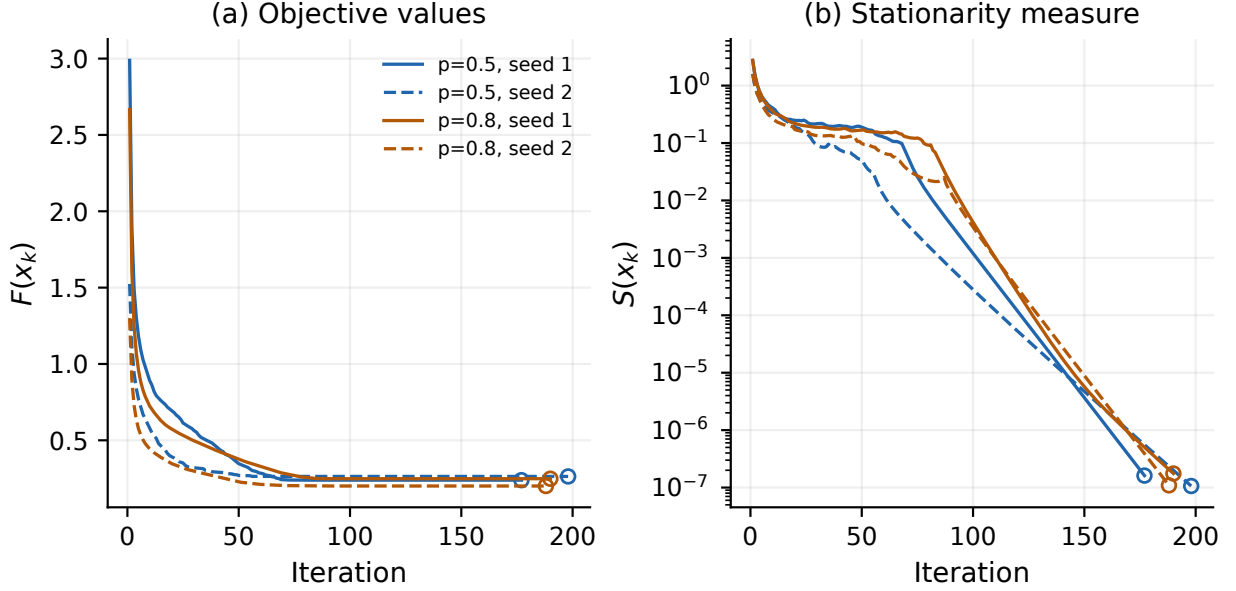


Figure 3: Objective values and stationarity measures for the series-based proximal-gradient runs. Solid and dashed curves distinguish the two instances; hollow circles show the corresponding Newton-based terminal values. The full traces for both evaluators are included in the data files.

Reproducibility. The accompanying files contain the inputs, high-precision reference values, raw numerical records, software versions, and scripts for reproducing the three studies and generating the figures and tables.

References

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